

KULLIYAH OF INFORMATION AND COMMUNICATION TECHNOLOGY

BICS 4340 – MACHINE LEARNING

Project Proposal

Topic:

Real-Time Bahasa Isyarat Malaysia (BIM) Recognition System Using YOLOv11 and MediaPipe
Keypoint Tracking for Deaf Community Accessibility

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Proposed New Research Title

Real-Time Bahasa Isyarat Malaysia (BIM) Recognition System Using YOLOv11 and MediaPipe Keypoint Tracking for Deaf Community Accessibility

Project Summary

Malaysia has its own unique sign language called **Bahasa Isyarat Malaysia (BIM)**, used by approximately 300,000+ Deaf and **Hard-of-Hearing (DHH)** individuals in the country. Unlike American Sign Language (ASL), BIM has its own unique hand gestures, grammar structure, and regional variations influenced by Malay culture and the Malaysian education system. Despite its importance, BIM remains severely underrepresented in AI-driven assistive technology research.

This proposed project adapts the same core pipeline from the referenced article, combining **YOLOv11** for gesture detection with **MediaPipe** for 21-keypoint hand tracking, but retrains and fine-tunes the model specifically on **BIM hand gestures**, covering the BIM alphabet and commonly used everyday signs. A large-scale BIM dataset would need to be curated from Malaysian signers across diverse demographics, skin tones, lighting conditions, and backgrounds, reflecting Malaysia's multicultural population. The system would be deployable via standard webcam on affordable hardware, making it accessible to schools, clinics, government offices, and public service counters across Malaysia.

Connection to SDGs (Sustainable Development Goals)

SDG 4 - Quality Education

BIM recognition technology can be integrated into Malaysian classrooms and e-learning platforms, enabling Deaf students in national schools and special education institutions (*Sekolah Pendidikan Khas*) to interact more effectively with teachers and learning content in real time, without waiting for a human interpreter.

SDG 3 - Good Health and Well-Being

In Malaysian hospitals and clinics under the Ministry of Health, communication between DHH patients and healthcare providers is a persistent problem. A real-time BIM interpreter embedded in hospital kiosks or tablets would help Deaf patients accurately convey symptoms and understand medical instructions, reducing the risk of poor health outcomes.

SDG 10 - Reduced Inequalities

The Malaysian DHH community faces significant social and economic exclusion. By providing an AI-driven BIM tool accessible through standard hardware, this project directly reduces the communication inequality gap, allowing Deaf individuals to participate more fully in public services, workplaces, and daily life.